

Reconstructing the Clebsch code from its deep-hole syndrome locus

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Abstract

Let A be a six-arc in $\text{PG}(2, 11)$, and let $\mathcal{U}(A)$ be the projective points lying on no chord of A . For the associated $[6, 3, 4]_{11}$ MDS code, $\mathcal{U}(A)$ is its projective deep-hole syndrome locus. We prove that $\mathcal{U}(A)$ lies on a conic if and only if A is projectively equivalent to the Clebsch hexagon; in that case $\mathcal{U}(A)$ is exactly a nonsingular conic. Thus nearest-codeword data reconstruct the non-GRS Clebsch code up to monomial equivalence. The reconstructed conic then determines its polarity, and Dye's theorem identifies the stabilizer as A_5 ; none of this geometry is assumed.

For the resulting code, the reconstructed conic equation is an exact syndrome-distance oracle, while decoder ambiguity recovers the Brianchon geometry and an invariant bipartition of the three-coordinate supports. The conceptual tools are a universal chord-defect identity and a partial-cover bound. Together with passant counts, they show that any k -arc whose uncovered locus is a nonsingular conic has q odd and

$$2k - 3 \leq q \leq \frac{k(k-1) + 3}{3}.$$

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1 Introduction

The complete nearest-codeword structure of a code can retain geometry that is absent from its definition. The example here is a non-GRS $[6, 3, 4]_{11}$ MDS code. Its maximum-distance syndrome directions form a conic, and that coarse locus already determines the projective code: its six parity-check columns are projectively equivalent to the Clebsch hexagon, with the associated polarity and A_5 symmetry. No conic, polarity, group action, or classical hexagon structure is assumed. Thus the problem is an inverse one: when does the projective deep-hole locus determine an MDS code, and how much of its geometry can nearest-codeword data recover?

Let C be a linear $[n, k, d]_q$ code with covering radius ρ and full-rank parity-check matrix H . For a received word v , write $\sigma(v) = Hv^\top$ and

$$w(s) = \min\{\text{wt}(e) : He^\top = s\}.$$

Then $d(v, C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ . The syndrome s specifies one coset of C , whose coset leaders are its weight- $w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\text{proj}}(C) = \{[s] \in \text{PG}(n - k - 1, q) : w(s) = \rho\}.$$

Thus one projective direction represents $q - 1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n - k$, the parity-check columns form a projective arc in $\text{PG}(r - 1, q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r - 1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. We write $\mathcal{U}(A)$ for this extension locus. If $\mathcal{U}(A) \neq \emptyset$, then the associated code has covering radius three and $\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(C)$. This dictionary is classical in the theory of complete arcs and is developed for MDS codes by Davydov, Marcugini, and Pambianco [8]; their Theorem 6.3 supplies the leader count used in Section 4. See also Hirschfeld [12] for the plane arc/covering-radius setting. We call A *conic-filling* when $\mathcal{U}(A)$ is the full $\mathbb{F}_q$ -rational point set of a nonsingular conic.

Theorem 1.1 (Rigidity theorem). *For a six-arc $A \subset \text{PG}(2, 11)$, the following are equivalent:*

- (i) *the uncovered locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);*
- (ii) *$\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$ -points of a nonsingular conic;*
- (iii) *A is $\text{PGL}(3, 11)$ -equivalent to the Clebsch hexagon.*

The proof reconstructs the hexagon from conic containment alone. A line bound, a chord-defect identity, and Dye’s Brianchon theorem give the conceptual implication. The same geometry then gives an exact distance oracle and all nearest-codeword multiplicities; the resulting ambiguity strata reconstruct the Brianchon points, the self-polar triangles, and the invariant support bipartition.

The chord-defect identity has a second role independent of the Clebsch classification. For a Clebsch hexagon K it gives $|\mathcal{U}(K)| = q^2 - 14q + 45$, with off-conic excess $(q - 4)(q - 11)$ when $q \equiv 3 \pmod{4}$. For an arbitrary conic-filling k -arc, the moment calculation and a partial-cover theorem force q odd and $2k - 3 \leq q \leq (k(k - 1) + 3)/3 < \binom{k}{2}$.

The configuration goes back to Clebsch [4, p. 336]. Edge constructed its finite-plane form and determined its order-60 stabilizer [5, §§29–32]; Dye proved projective uniqueness and determined the stabilizer over a general field [10, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [17]. Deep holes of Reed–Solomon codes have both an algorithmic history [11, 7] and a geometric classification in the projective case [18]. Those results concern GRS codes, whose arcs lie on normal rational curves; the code here is not GRS.

What is new and what is not. The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [4, 5, 10, 2, 17]. We prove the conic-locus identification of Proposition 4.2 directly here. The new contribution is the symmetry-free coding criterion, the decoder reconstruction and support bipartition, and the uniform conic-filling window. A separate computational companion [15] proves sharper finite consequences: a degree-three characterization and numerical gap at $q = 11$, uniqueness of $q = 11$ among conic-filling six-arcs, and classification through eight points.

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C} : XZ = Y^2$$

in $\text{PG}(2, 11)$, with its 12 points identified with $\text{PG}(1, 11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\text{PGL}(3, 11)$ is $\text{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

A convenient representative is

$$A = \{(1, 10, 0), (1, 9, 1), (1, 4, 7), (1, 8, 5), (0, 1, 4), (1, 1, 7)\}.$$

It is a six-arc disjoint from $\mathcal{C}$; its fifteen joins also avoid $\mathcal{C}$. The following invariant construction explains its symmetry.

Definition 2.1 (Clebsch hexagon). Let $A_5 \subset \text{PGL}_2(11)$ be an icosahedral subgroup. The six Sylow 5-subgroups of A_5 each fix two points of $\mathcal{C}$. The *Clebsch hexagon* is the set of poles of the six chords joining these fixed pairs. More generally, following Dye, a Clebsch hexagon over a field is a six-arc with exactly ten nonvertex triple-chord concurrence points, called its Brianchon points.

For a nonsingular conic over an odd field, an *external point* lies on two tangents and an *internal point* on none; an *exterior*, or *passant*, line is disjoint from the conic. Edge's six vertices are external, their fifteen joins are exterior, and those joins concur three at a time at ten internal Brianchon points [5, §§29–32]. Thus the hexagon is a complete exterior set in the terminology of Blokhuis, Seress, and Wilbrink [2, pp. 143,146]. Their $q = 11$ list also contains a Pasch configuration, but its four collinear triples prevent it from defining an MDS code. Thus exteriority alone gives only $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; the arc hypothesis selects the Clebsch branch, and Proposition 4.2 proves equality.

The displayed hexagon is the arc K_2 of Storme and Van Maldeghem [17, Props. 11–12] and Dye's synthetic hexagon when 5 is a square in the base field [10, Theorem 1, pp. 275–276]. In characteristic 11, Dye's Theorems 1–3 give projective uniqueness, the unique associated polarity, and stabilizer A_5 ; in the displayed coordinates the associated conic is $\mathcal{C}$ [10, pp. 275–278]. His calculation also shows that the six vertices lie on no conic in characteristic 11 [10, Theorem 2, pp. 277–278; p. 281]. Storme and Van Maldeghem record that the arc is incomplete at $q = 11$ [17, Prop. 13], so the associated redundancy-three code has covering radius three.

3 A universal chord-defect identity

Theorem 3.1 (Universal chord defect). *Let A be a k -arc in $\text{PG}(2, q)$ with $k \geq 4$. Let n_i count the off-arc points incident with exactly i chords, put*

$$r = \lfloor k/2 \rfloor, \quad t(A) = \sum_{i=3}^r \binom{i-1}{2} n_i,$$

and write $m = \binom{k}{2}$. Then

$$|\mathcal{U}(A)| = q^2 - (m-1)q + \left(m + 3 \binom{k}{4} - k + 1 \right) - t(A),$$

$$0 \leq t(A) \leq 3 \binom{k}{4} \frac{r-2}{r}, \quad k \leq 5 \implies t(A) = 0.$$

Proof. At a point off A , concurrent chords have pairwise disjoint endpoint pairs, so at most r chords occur. Counting chord–point incidences and pairs of disjoint chords gives

$$\sum_i in_i = m(q-1), \quad \sum_i \binom{i}{2} n_i = 3 \binom{k}{4}.$$

Since $\binom{i}{2} = (i-1) + \binom{i-1}{2}$, the number of covered off-arc points is $m(q-1) - 3\binom{k}{4} + t(A)$. Subtraction from the $q^2 + q + 1 - k$ off-arc points proves the identity. Finally,

$$\binom{i-1}{2} \leq \frac{r-2}{r} \binom{i}{2} \quad (3 \leq i \leq r),$$

and the second moment gives the defect bound. $\square$

Corollary 3.2 (Conic-filling window and six-arcs). *If $|\mathcal{U}(A)| = q + 1$, then*

$$t(A) = q^2 - mq + m + 3 \binom{k}{4} - k, \quad q^2 - mq + m - k + \frac{6}{r} \binom{k}{4} \leq 0,$$

and $q < m$. Under the same hypothesis $|\mathcal{U}(A)| = q + 1$, if q is even, then $\mathcal{U}(A)$ is not an arc. Hence if $\mathcal{U}(A)$ is a nonsingular conic, then q is odd and

$$2k - 3 \leq q \leq \frac{k(k-1) + 3}{3} < \binom{k}{2}.$$

For $k = 6$, writing $c(A) = n_3$ gives

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \quad 0 \leq c(A) \leq 15, \quad c(A) = (q-6)(q-9)$$

when $|\mathcal{U}(A)| = q + 1$. For the displayed Clebsch hexagon, Dye’s ten Brianchon points give $c(A) = 10$; its fifteen chords have 45 disjoint pairs, cover 115 of the 127 off-arc points, and leave exactly twelve uncovered.

Proof. The first formula is Theorem 3.1 rearranged, and its comparison with the defect bound gives the quadratic inequality. There are $q^2 - k$ covered off-arc points, so $q^2 - k \leq m(q-1)$; if $q \geq m$, then $0 \leq q(q-m) \leq k-m < 0$, a contradiction.

Suppose q is even and $\mathcal{U}(A)$ is an arc. Since it has $q+1$ points, it has a nucleus $N \notin \mathcal{U}(A)$ through which every line is tangent [12, p. 177]. Every chord of A is disjoint from $\mathcal{U}(A)$, so no chord contains N ; nor can N be a vertex. Thus $N \in \mathcal{U}(A)$, a contradiction.

In the conic case, q is therefore odd and every chord is passant. An off-conic point lies on at most $(q+1)/2$ passants, so the $k-1$ chords through a vertex give $q \geq 2k-3$.

The m chords form a partial line cover whose $q+1$ holes are the noncollinear points of the conic. The partial-cover bound of Blokhuis, Brouwer, and Szőnyi gives [1, Proposition 1.5]

$$m \geq 2q - 1 - \frac{q+1}{2} = \frac{3(q-1)}{2}.$$

Rearranging gives $q \leq (k(k-1) + 3)/3$; the final strict inequality follows from $k \geq 4$.

For $k = 6$, one has $r = 3$ and $t(A) = n_3 = c(A)$, giving the displayed specialization. For the displayed arc, Dye’s Brianchon theorem gives $c(A) = 10$ [10, Theorem 1, pp. 275–276]. At $q = 11$, the two moments read $\sum in_i = 150$ and $\sum \binom{i}{2} n_i = 45$, whence $n_1 + n_2 + n_3 = 115$. $\square$

4 The code and its projective syndrome locus

Take the parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 10 & 9 & 4 & 8 & 1 & 1 \\ 0 & 1 & 7 & 5 & 4 & 7 \end{pmatrix},$$

whose columns are the six displayed points of A , and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^\top = 0\}$$

be the resulting $[6, 3, 4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [16].

The arc-coset dictionary from the introduction applies without a GRS hypothesis [8, Def. 6.1, Thm. 6.3]. Each uncovered direction x lifts to $q - 1$ weight-three cosets, and each has $\binom{6}{3} = 20$ minimum-weight leaders. Geometrically, the same condition says that adjoining x as a seventh column preserves the MDS property.

For example, $s = (1, 0, 0)$ lies on $\mathcal{C}$ and is therefore uncovered. The error

$$e = (7, 4, 1, 0, 0, 0)$$

satisfies $He^\top = s$. No error of weight one or two has syndrome s , while the twenty triples of coordinate positions each supports one weight-three leader. Thus this single syndrome exhibits the complete dictionary: its projective direction is a conic point, its coset has distance three, and its nearest-codeword ambiguity has size twenty.

The stabilizer already forces the small point sets that arise below.

Proposition 4.1 (The A_5 point orbits). *Let $G = \text{Stab}(A) \cong A_5$. Its orbits on $\text{PG}(2, 11)$ have lengths*

$$6, 10, 12, 15, 30, 30, 30.$$

The first four are respectively A , the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G -orbit.

Proof. Dye's polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [10, Theorems 2–3 and Corollary 1, pp. 276–279]. We derive the orbit sizes without using the finite enumeration.

The argument first determines the points with each possible nontrivial stabilizer, then applies orbit-stabilizer to recover the orbit lengths.

Since $11 \nmid 60$, the representation is semisimple, and the ternary icosahedral representation is irreducible; hence G fixes no projective point. The ordinary three-dimensional icosahedral character, reduced in $\mathbb{F}_{11}$, shows that an involution, an element of order three, and an element of order five fix respectively 13, 1, and 3 projective points. Indeed, an involution has a one-dimensional eigenspace and a two-dimensional eigenspace; the nontrivial order-three eigenvalues are not in $\mathbb{F}_{11}$; and the three order-five eigenvalues are distinct because $5 \mid 10$. Restricting the same representation to the standard subgroups of A_5 gives the following common-fixed-point ledger; the standard subgroup classification of A_5 shows

that these are all possible nontrivial proper point stabilizers. Here D_5 has order ten, and the last row counts points whose stabilizer is exactly the displayed subgroup type. For the noncyclic entries, determine the common projective eigenlines of standard generators: a D_5 normalizer retains one of the three C_5 eigenlines, an S_3 normalizer retains the unique C_3 eigenline, a V_4 has three simultaneous eigenlines, and its overgroup A_4 permutes those three and fixes none. This gives the second row.

H	A_4	D_5	S_3	C_5	V_4	C_3
number of conjugate subgroups	5	6	10	6	5	10
fixed points for one H	0	1	1	3	3	1
points with stabilizer conjugate to H	0	6	10	12	15	0

For the third row, a D_5 fixes one C_5 eigenline and exchanges the other two; the unique C_3 line is already fixed by its S_3 normalizer; and A_4 permutes the three V_4 eigenlines without fixing one. By orbit–stabilizer, the third row already gives the orbits of lengths 6, 10, 12, and 15. To account for the points with exact stabilizer C_2 , observe that each involution lies in two D_5 subgroups, two S_3 subgroups, and one V_4 . Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer C_2 . There are fifteen involutions, so this gives 90 points, necessarily three orbits of length $60/2 = 30$. The total $6 + 10 + 12 + 15 + 90 = 133$ exhausts the plane. Dye’s Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point C_5 -stabilizer orbit with $\mathcal{C}(\mathbb{F}_{11})$ [10, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above. $\square$

Proposition 4.2 (The uncovered locus is the conic). *The covering radius of C is three, and*

$$\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(C) = \mathcal{C}(\mathbb{F}_{11}).$$

Proof. Corollary 3.2 gives $|\mathcal{U}(A)| = 12$. The exteriority of all fifteen chords gives $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$. Since a nonsingular conic has twelve $\mathbb{F}_{11}$ -points, equality follows without using the orbit classification. The arc–coset dictionary identifies this set with the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three. $\square$

Corollary 4.3 (Directions, cosets, leaders, and received words). *$\mathcal{U}(A)$ has 12 directions, exactly the points of C . Above them lie $12(11 - 1) = 120$ nonzero maximum-distance syndromes, hence 120 deep-hole cosets. Each such coset has $\binom{6}{3} = 20$ minimum-weight leaders, for 2400 leaders in all, and contains $|C| = 11^3 = 1331$ received words. Consequently C has $120 \cdot 1331 = 159720$ received-word deep holes.*

Proof. Immediate from Proposition 4.2, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \rightarrow \text{PG}(2, 11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. $\square$

4.1 Automorphisms and deep-hole orbits

The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the coordinate supports. It is the support-permutation quotient of the monomial automorphism group:

$$1 \longrightarrow \mathbb{F}_{11}^\times \longrightarrow \text{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars: a projectivity fixing all six column rays fixes a projective frame and is the identity. The sequence splits. Since $\gcd(3, 10) = 1$,

each projective class has a unique determinant-one representative. These representatives are closed under multiplication, and their induced monomial lifts are unique once the displayed column representatives are fixed; hence they give an A_5 complement. The global scalars are central, so

$$\text{MAut}(C) \cong C_{10} \times A_5$$

has order 600.

Proposition 4.4 (One orbit on received-word deep holes). *The monomial automorphism group of C is transitive on the 120 deep-hole cosets. Let Γ be the group generated by these automorphisms and by translations $v \mapsto v + c$ for $c \in C$. Then*

$$|\Gamma| = |C| |\text{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six fixed-point chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^\times$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w , choose $m \in \text{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w . Both kinds of generator are Hamming isometries preserving C . The monomial group normalizes the translation subgroup, since $m t_c m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \text{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. $\square$

4.2 Decoding consequences

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight 0, 1, 2, 3, summing to $11^3 = 1331$); the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters. This is the radius-three instance of the coset-weight framework of Davydov, Marcugini, and Pambianco [8, §6]; the secant indices below refine its weight-two stratum by nearest-codeword multiplicity.

The special geometry also makes distance diagnosis and nearest-codeword structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$. For a point $x \notin A$, its *secant index* is the number of chords of A through x ; by the dictionary above, this is the number of weight-two leaders for each nonzero syndrome on the ray x .

Proposition 4.5 (Complete syndrome oracle and ambiguity enumerator). *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = H v^\top$,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

Among the 1330 nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are

$$960, \quad 150, \quad 100, \quad 120.$$

For a distance-two syndrome the number of nearest codewords is the secant index of its direction. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 4.2 identifies the last set with $Q = 0$, proving the oracle. In the notation of its proof, the chord-intersection equations give $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$, where n_i counts projective directions of secant index i and n_0 is the uncovered locus. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. $\square$

4.3 Self-polar structure

Edge's finite-plane construction [5, §§30.2–31] and Dye's general-field synthetic theory [10, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for $\mathcal{C}$. Their vertex-pair matchings partition the fifteen pairs into five perfect matchings of K_6 . Section 6 turns its ten pairs of matchings into the Brianchon–support dictionary.

5 The rigidity theorem

Proposition 4.2 concerns a specific arc. We now prove the symmetry-free characterization stated in Theorem 1.1. The uncovered conic determines the resulting Clebsch hexagon's polarity, and Dye's stabilizer theorem then identifies its projective stabilizer as A_5 ; neither is assumed.

Conic containment. The condition is that the uncovered locus $\mathcal{U}(A)$ lies on a conic, not that the arc A itself does. The proof below uses only the former condition and imposes no hypothesis that the six vertices lie on a conic.

We first isolate the geometric input needed to exclude degenerate containing conics.

The exterior-line case is a six-point instance of the nonexistence in odd order of nontrivial *hyperfocused arcs*, whose secants determine only $k - 1$ points on an exterior line [3, 6, 14]. We give a direct proof because the three possible line positions are then handled together.

Lemma 5.1 (Line bound). *Let A be a six-arc in $\text{PG}(2, q)$, where q is odd. Every line contains at most $q - 5$ points of $\mathcal{U}(A)$.*

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points. Indeed, two of those chords sharing an endpoint meet only at that endpoint, which lies off ℓ ; their intersections with ℓ therefore give a proper edge-colouring of K_5 , whose chromatic index is five. Together with the vertex, at least six of the $q + 1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \emptyset$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them: chords sharing an endpoint meet only at that arc vertex, not on ℓ , so all chords through one point of ℓ have disjoint endpoint pairs. Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6 , necessarily a one-factorization. The one-factorization of K_6 is unique up to isomorphism [13]; after relabelling, three factors are

$$01|23|45, \quad 02|14|35, \quad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0 = (0, 0)$, $p_1 = (x, 0)$ and $p_2 = (0, y)$, where $xy \neq 0$. Writing $p_3 = (a, y)$, the first two factors force $p_4 = (x, b)$ and $p_5 = (a, b)$. Parallelism of 03 with 15 and with 24 gives

$$ab - ay + xy = 0, \quad ab - ay - xy = 0.$$

Their difference is $2xy = 0$, impossible for odd q . Thus a disjoint line has at least six covered points, and in all cases at most $(q + 1) - 6 = q - 5$ points of $\mathcal{U}(A)$ remain. $\square$

Proof of Theorem 1.1. Suppose first that $\mathcal{U}(A)$ lies on a conic Q . If Q is nonsingular then $|\mathcal{U}(A)| \leq q + 1 = 12$. If Q is degenerate, its rational points lie on one or two rational lines when it splits; in the nonsplit rank-two case its only rational point is the singular point. Thus Lemma 5.1 again gives $|\mathcal{U}(A)| \leq 12$. On the other hand, the chord-defect identity of Theorem 3.1 says

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ is the number of nonvertex triple-edge concurrences. In Dye's terminology these are precisely the Brianchon points of the hexagon A . Over a field of characteristic different from two, Dye proves that every six-arc has at most ten Brianchon points and that the equality cases form a single $\mathrm{PGL}_3(K)$ -orbit [10, §2.2–2.3, Theorem 1, pp. 275–276]. This bound has no symmetry, conic, or genericity hypothesis beyond the six-arc assumption. Hence $c(A) \leq 10$, so $|\mathcal{U}(A)| \geq 12$; equality holds throughout, $c(A) = 10$, and A is a Clebsch hexagon.

The degenerate-conic branch is therefore vacuous. Indeed, Proposition 4.2 and projective equivariance identify $\mathcal{U}(A)$ with the twelve rational points of a nonsingular conic. A degenerate conic cannot contain those same twelve points: it has no component in common with the nonsingular conic, so Bézout's theorem bounds their intersection by four. Thus the original containing conic is nonsingular. This proves (i) $\Rightarrow$ (iii). Proposition 4.2 and projective equivariance give (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i). Dye's stabilizer theorem also identifies the stabilizer in these equivalent cases with A_5 . $\square$

Projective equivalence of the unordered parity-check column sets is equivalent to monomial equivalence of the associated codes: choose column representatives after the projectivity, then absorb their nonzero scalar factors into coordinate scalings. Thus Theorem 1.1 also proves the coding formulation in the abstract.

Remark 5.2 (Why this rigidity is not in the extension-locus literature). The set $\mathcal{U}(A)$ is the classical extension locus, studied for sufficiently large arcs through Segre's tangent-envelope method; see Ball and Lavrauw [9, Thms. 39–41]. Those results require arc-size hypotheses far from the six-point regime and do not record the conic rigidity detected here.

6 The invariant support bipartition

Label the six coordinates by $0, \dots, 5$. The 20 three-element coordinate supports form two complementary A_5 -orbits of size ten. Automorphisms preserve the two classes, but nothing intrinsic chooses one as positive. We use *support bipartition* only as shorthand for this unoriented bipartition. Its ten complementary pairs carry the Petersen graph in Figure 1.

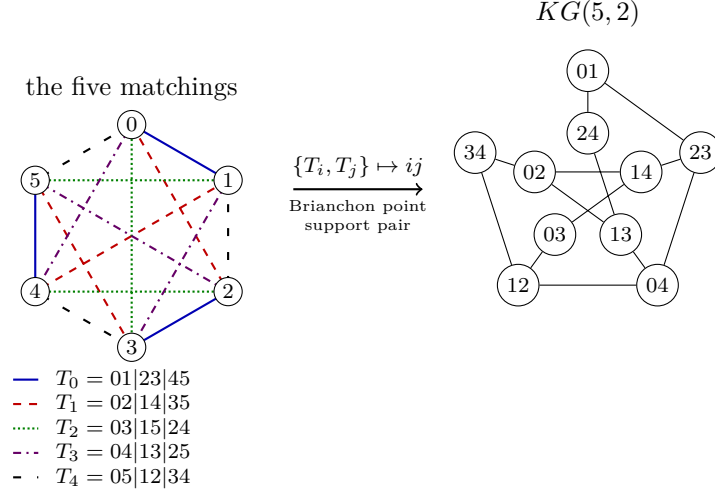


Figure 1: The five-matchings–Petersen dictionary. The line styles encode the five self-polar triangles as perfect matchings of the six coordinates. For each pair $\{T_i, T_j\}$, their union is an alternating six-cycle: its opposite-edge matching gives a Brianchon point, and its alternating vertex classes give a complementary support pair. Both are represented by the Petersen vertex ij ; adjacency means that the index pairs are disjoint.

Proposition 6.1 (Brianchon–support dictionary). *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [5, 10], regarded as five perfect matchings, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T, T')$ be the perfect matching joining opposite vertices of this cycle. Let $\beta(T, T') = \{S, S^c\}$ be the unordered pair of alternating vertex classes of the cycle. For example, $T_0 \cup T_1$ is the cycle $0, 1, 4, 5, 3, 2$, so $M(T_0, T_1) = 05|13|24$ and $\beta(T_0, T_1) = \{\{0, 3, 4\}, \{1, 2, 5\}\}$.*

The three chords indexed by $M(T, T')$ concur at a point $b(T, T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \quad \{T, T'\} \longmapsto b(T, T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S, S^c\} : |S| = 3\}, \quad \{T, T'\} \longmapsto \beta(T, T'),$$

are A_5 -equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of $\mathcal{T}$. Hence $\{T, T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T, T')$, so the ten triangle pairs give exactly the ten perfect matchings outside $\mathcal{T}$. Applying the alternating-class rule to the five displayed matchings gives ten

distinct complementary support pairs, hence all ten such pairs. Edge's Clebsch construction says that the three chords of each such matching concur at the ten distinct Brianchon points [5, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above.

The ten outside matchings give the ten triple concurrences. The remaining fifteen intersections of disjoint chords have multiplicity one. Since every step uses only the five invariant matchings and incidence, both bijections and their compatibility with the support map are A_5 -equivariant. $\square$

Corollary 6.2 (The decoder reconstructs the Brianchon configuration). *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five matchings in $\mathcal{T}$.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 6.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings outside $\mathcal{T}$. $\square$

Proposition 6.3 (Invariant support bipartition). *Only the bipartition is intrinsic: neither part has a canonical sign. The twenty three-element coordinate supports split into two complementary A_5 -orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s , each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as $1200 + 1200$.*

Every monomial automorphism of C preserves this unordered bipartition. The nontrivial coset of A_5 in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. By Proposition 6.1, the ten complementary support pairs are canonically identified both with the two-subsets of the five self-polar triangles and with the ten Brianchon points.

Proof. The degree-six action of A_5 on the twenty three-subsets has two ten-element orbits, exchanged by complementation. It preserves a unique set of five perfect matchings, also preserved by its full S_5 normalizer. The alternating-cycle construction of Figure 1 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a maximum-distance syndrome s and a three-support S . The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s . None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A , contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S . Finally, every monomial automorphism induces a support permutation in A_5 , by the exact sequence in Section 4.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits but cannot preserve C , even after coordinate rescaling, because its support permutations lie outside that quotient A_5 . $\square$

7 The Clebsch family across fields

Recall that A is conic-filling when $\mathcal{U}(A) = \mathcal{Q}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{Q}$. Equivalently, $A \cap \mathcal{Q}(\mathbb{F}_q) = \emptyset$, its chords cover every point outside $A \cup \mathcal{Q}(\mathbb{F}_q)$, and no chord meets $\mathcal{Q}(\mathbb{F}_q)$. For the Clebsch family, the chord-defect identity gives a direct formula for the uncovered locus and isolates the conic-filling specialization.

Proposition 7.1 (The Clebsch-family uncovered formula). *Let K be a Clebsch hexagon over $\mathbb{F}_q$; equivalently, K is a six-arc with ten Brianchon points. By Dye’s existence theorem, the characteristic is not two and 5 has a square root in $\mathbb{F}_q$, possibly zero in characteristic five. Then*

$$|\mathcal{U}(K)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod{4}$ and $\mathcal{C}$ is Dye’s associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(K)$ and

$$|\mathcal{U}(K) \setminus \mathcal{C}(\mathbb{F}_q)| = (q - 4)(q - 11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q = 11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [10, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(K)$. Thus $c(K) = 10$, and Theorem 3.1 gives

$$|\mathcal{U}(K)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of K are non-secants of the associated conic when $q \equiv 3 \pmod{4}$ [10, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q - 4)(q - 11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye’s Theorem 1, and the inclusion above is then an equality because both sets have twelve points. $\square$

Remark 7.2 (Three Clebsch-family regimes). The polynomial $q^2 - 14q + 45 = (q - 5)(q - 9)$ vanishes exactly at $q = 5, 9$. The root $q = 5$ lies in characteristic five, where the associated-conic construction of Definition 2.1 is not available. At $q = 9$, however, Dye’s construction applies: $5 = -1$ is a square in $\mathbb{F}_9$, and the characteristic is three. Thus the formula gives an empty uncovered locus, so the Clebsch six-arc is complete and not conic-filling. At $q = 11$ the uncovered locus is instead exactly the associated conic.

At $q = 19$, the same icosahedral construction still produces a Clebsch six-arc disjoint from its associated conic, but the covering fails. The twenty conic points remain uncovered [10, p. 281], while Proposition 7.1 gives

$$|\mathcal{U}(A)| = 140 = 20 + 120.$$

Thus $120 = (19 - 4)(19 - 11)$ additional points escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic. The configuration persists; exact conic filling is what the factorization isolates at $q = 11$.

8 Proof sources and formal cross-check

The rigidity and field-window arguments use three external mathematical inputs. Dye’s Brianchon bound and equality classification identify the Clebsch orbit [10, §2.2–2.3, Theorem 1, pp. 275–276]; his stabilizer and associated-conic results supply the classical Clebsch geometry [10, Theorems 2–3, pp. 276–278]. The even-order oval obstruction uses the standard nucleus theorem [12, p. 177]. The upper conic-filling bound is Proposition 1.5 of Blokhuis, Brouwer, and Szőnyi [1].

The line bound, chord-defect identity, partial-cover deduction, decoder reconstruction, and support bipartition are proved in the text. A separate Lean development checks the central deductions conditional on the two Dye inputs. It is distributed in the `finitegeom` repository at commit `6d4766d1ea5e9a36f1a507e549c223416a6b506f`. The aggregate theorem is `RelativeConicArcs/Gates/ClebschRigidityTrust.lean`; the development uses Lean 4.32.0-rc1 and Mathlib commit `571b8a8e54219b4d393f75f4b8653fac08197fcc`. This formalization is an independent cross-check, not a proof dependency of the manuscript. Its recorded axiom audit lists the standard logical axioms and, for the rigidity implication, exactly the explicit Brianchon-bound and equality-classification axioms imported from Dye; the gate uses neither admitted declarations nor native execution.

9 Conclusion

For every fixed arc size, the partial-cover bound limits the field orders that can admit conic filling. Together with the universal defect identity and passant counts, it gives

$$2k - 3 \leq q \leq \frac{k(k - 1) + 3}{3}.$$

The Clebsch code realizes a different kind of sharpness: covering-radius data recover geometry that was not built into the code, forcing the Clebsch hexagon and hence its A_5 symmetry. A line bound and the defect identity reduce that rigidity to Dye’s equality case.

More generally, which odd prime powers in the quadratically widening window admit conic-filling arcs? A second problem is to connect fixed- k rigidity with the large-arc conic-forcing regime. Both ask how far a syndrome locus can reconstruct the projective system that produced it.

The same questions have useful algebraic and spectral forms. Passant conditions become clique problems in conic association schemes, while the decoder results ask when coset-leader support data alone reconstruct a projective code.

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